9. Python Program to Implement the Travelling Salesman Problem

Aim: To implement the Travelling Salesman Problem (TSP) in Python and find the minimum-cost tour that visits every city once and returns to the starting city.

Procedure:

1. Define the number of cities and their travel-cost matrix.
2. Fix the first city as the starting point.
3. Generate all possible orders of visiting the remaining cities.
4. Calculate the total travel cost for each possible tour.
5. Select the tour with the minimum total cost.
6. Display the optimal tour and its minimum cost.

Code:

```
from itertools import permutations

cost = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

n = len(cost)
cities = range(1, n)
min_cost = float('inf')
best_path = []

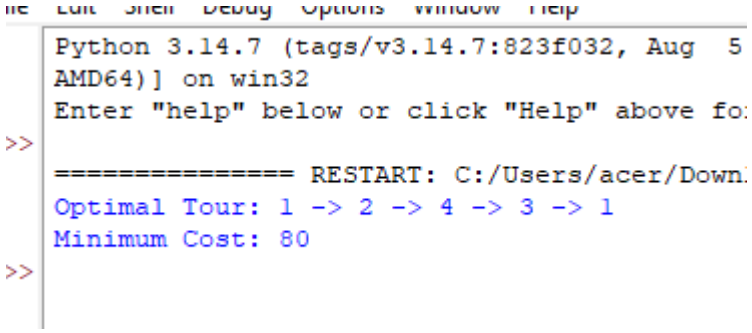
for path in permutations(cities):
    tour = (0,) + path + (0,)
    total = 0

    for i in range(len(tour) - 1):
        total += cost[tour[i]][tour[i + 1]]

    if total < min_cost:
        min_cost = total
        best_path = tour

print("Optimal Tour:", " -> ".join(
    str(city + 1) for city in best_path))
print("Minimum Cost:", min_cost)
```

Output:



Result: Thus, the Travelling Salesman Problem was implemented successfully using Python, and the minimum-cost tour was found.